

Quantum Computing, Quantum Machine Learning and Quantum Information Theories

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Plans for the week of February 17-21, 2025

The first part of the lecture will focus

1. Review from last week, one-qubit gates and one-qubit Hamiltonian (using slides from last week)
2. Reminder and review of density matrices and measurements from week 2
3. Schmidt decomposition and entanglement
4. Discussion of entropies, classical information entropy (Shannon entropy) and von Neumann entropy

The second part focuses on

1. Two-qubit Hamiltonians and how to encode them
2. Discussion of project 1 during exercise session

Motivation

In order to study entanglement and why it is so important for quantum computing, we need to introduce some basic measures and useful quantities. For these endeavors, we will use our two-qubit system from the second lecture in order to introduce and repeat, through examples, density matrices and entropy. These two quantities, together with technicalities like the Schmidt decomposition define important quantities in analyzing quantum computing examples.

The Schmidt decomposition is again a linear decomposition which allows us to express a vector in terms of tensor product of two inner product spaces. In quantum information theory and quantum computing it is widely used as a way to define and describe entanglement.

Reminder on density matrices and traces

We have the spectral decomposition of a given operator $\mathbf{A}$ given by

$$\mathbf{A} = \sum_i \lambda_i |i\rangle\langle i|,$$

with the ONB $|i\rangle$ being eigenvectors of $\mathbf{A}$ and λ_i being the eigenvalues. Similarly, a operator which is a function of $\mathbf{A}$ is given by

$$f(\mathbf{A}) = \sum_i f(\lambda_i) |i\rangle\langle i|.$$

The trace of a product of matrices is cyclic, that is

$$\mathrm{tr}[\mathbf{ABC}] = \mathrm{tr}[\mathbf{BCA}] = \mathrm{tr}[\mathbf{CBA}],$$

and we have also

$$\mathrm{tr}[\mathbf{A}|\psi\rangle\langle\psi|] = \langle\psi|\mathbf{A}|\psi\rangle.$$

Definition of density matrix

Using the spectral decomposition we defined also the density matrix as

$$\rho = \sum_i p_i |i\rangle\langle i|,$$

where the probability p_i are the eigenvalues of the density linked with the ONB $|i\rangle$.

The trace of the density matrix is $\text{tr}\rho = 1$ and it is invariant under unitary transformations $|\psi'_i\rangle = \mathbf{U}|\psi_i\rangle$. The unitary transformation of the density matrix gives, with $\mathbf{U}^\dagger \mathbf{U} = \mathbf{U}^T \mathbf{U} = \mathbf{I}$,

$$\mathbf{U}\rho\mathbf{U}^\dagger = \sum_i p_i \mathbf{U}|\psi_i\rangle\langle\psi_i|\mathbf{U}^\dagger,$$

and with the unitary transformation it is easy to show that the trace of the transformed density matrix is equal to one,

$$\text{tr} [\mathbf{U}\rho\mathbf{U}^\dagger] = \text{tr} [\mathbf{U}\mathbf{U}^\dagger\rho] = 1.$$

Revisiting our Bell states and our entanglement encounter, two qubit system

We define a system that can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \quad |1\rangle_{A,B} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry.

Computational basis

This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \ 0 \ 0 \ 0]^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \ 1 \ 0 \ 0]^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \ 0 \ 1 \ 0]^T,$$

and finally

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \ 0 \ 0 \ 1]^T.$$

Bell states

The above computational basis states, which define an ONB, can in turn be used to define another ONB. As an example, consider the so-called Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

The next two

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

It is easy to convince oneself that these states also form an orthonormal basis.

Measurement

Measuring one of the qubits of one of the above Bell states, automatically determines, as we will see below, the state of the second qubit. To convince ourselves about this, let us assume we perform a measurement on the qubit in system A by introducing the projections with outcomes 0 or 1 as

$$P_0 = |0\rangle\langle 0|_A \otimes I_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

for the projection of the $|0\rangle$ state in system A and similarly

$$P_1 = |1\rangle\langle 1|_A \otimes I_B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for the projection of the $|1\rangle$ state in system A .

Probability of outcome

We can then calculate the probability for the various outcomes by computing for example the probability for measuring qubit 0

$$\langle \Phi^+ | \mathbf{P}_0 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

Similarly, we obtain

$$\langle \Phi^+ | \mathbf{P}_1 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

States after measurement

After the above measurements the system is in the states

$$|\Phi'_0\rangle = \sqrt{2} [|0\rangle\langle 0|_A \otimes I_B] |\Phi^+\rangle = |00\rangle,$$

and

$$|\Phi'_1\rangle = \sqrt{2} [|1\rangle\langle 1|_A \otimes I_B] |\Phi^+\rangle = |11\rangle.$$

We see from the last two equations that the state of the second qubit is determined even though the measurement has only taken place locally on system A .

Other states

If we on the other hand consider a state like

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B,$$

this is a pure **product** state of the single-qubit, or single-particle states, of two qubits (particles) in system A and system B , respectively. We call such a state for a **pure product state**.

Quantum states that cannot be written as a mixture of other states are called pure quantum states or just product states, while all other states are called mixed quantum states.

More on Bell states

A state like one of the Bell states (where we introduce the subscript AB to indicate that the state is composed of single states from two subsystem)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle_{AB} + |11\rangle_{AB}],$$

is on the other hand a mixed state and we cannot determine whether system A is in a state 0 or 1. The above state is a superposition of the states $|00\rangle_{AB}$ and $|11\rangle_{AB}$ and it is not possible to determine individual states of systems A and B , respectively.

Entanglement

We say that the state is entangled. This yields the following definition of entangled states: a pure bipartite state $|\psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choice of the states $|\psi\rangle_A$ and $|\phi\rangle_B$. Otherwise we say the state is separable.

Maximally entangled

A so-called maximally entangled state for a bipartite system has equal probability amplitudes

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle.$$

We call a bipartite state composed of systems A and B (these systems can be single-particle systems, or single-qubit systems representing low-lying states of complicated many-body systems) for separable if its density matrix ρ_{AB} can be written out as the tensor product of the individual density matrices ρ_A and ρ_B , that is we have for a given probability distribution p_i

$$\rho_{AB} = \sum_i p_i \rho_A(i) \otimes \rho_B(i).$$

Entropies and density matrices

We discuss first the classical information entropy. Thereafter we define the quantum-mechanical quantity, commonly known as the Von-Neumann entropy.

Shannon information entropy

We start our discussions with the classical information entropy, or just Shannon entropy, before we move over to a quantum mechanical way to define the entropy based on the density matrices discussed earlier.

We define a set of random variables $X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the classical information entropy is defined as

$$S = - \sum_{x \in X} p_X(x) \log_2 p_X(x).$$

Why this expression? What does it mean?

Mathematics of entropy

What is the basic idea of the entropy as it is used in information theory (we leave out the standard description from statistical physics here)?

We want to have a measure of unlikelihood. Consider a simple binary system, with two outcomes, true and false with true given by a probability p and false given by $1 - p$. Since p represents a probability, and assuming the probabilities are properly normalized, we have $p \in [0, 1]$.

Suppose now that we know precisely p , for example we could set $p = 1$. This means that there is only one outcome, namely the true one and false is equal to zero. We know thus (exactly) which state the system is in. If we set $p = 0$, then we know that the outcome is false, without doubt. Can we find a way to, through a specific function to say with a given certainty that the system is in a specific state? Let us call this function for S .

More on the mathematics of entropy

We could for example use

$$S(x) = -\log_2 p(x),$$

as potential function. Note that if we change the log-base, we only change the results by a scaling constant.

When $p(x) = 1$, then $S(x) = 0$ and we could say that the probability of being in one specific state is uniquely defined since S is then zero. This function is also continuous, it is additive (which is very useful if we have independent and identically distributed stochastic variables), it is also high for unlikely events since when p goes to zero (unlikely event) it becomes very large. It is also a non-negative function.

Changing the expression

We note however that if we go back to our binary model and would like to describe the entropy in terms of the variable p , we end up with p being zero for the false outcome. The latter has probability one ($p = 0$). Using that

$$\lim_{x \rightarrow -\infty} -x \log_2 x = 0,$$

we can define the classical information entropy as

$$S(x) = -p(x) \log_2 p(x).$$

If we now consider a series of stochastic variables

$X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the entropy becomes

$$S_X = - \sum_{x \in X} p_X(x_i) \log_2 p_X(x_i).$$

Binary example

If we now use the above binary example, we have (using that $p_1 = p$ and $p_2 = 1 - p$)

$$S = - \sum_{i=1}^2 p_i \log_2 p_i = -p \log_2 p - (1 - p) \log_2 1 - p.$$

For $p = 0$ we have $S = 0$ and for $p = 1$ we get $S = 0$. Thus, for the two uniquely defined states we have entropy zero and a precise knowledge of the state. The maximum value is at $p = 0.5$ and this is a situation where our level of knowledge is no better than us tossing a coin. This simple example demonstrates how we can use the above mathematical expression as a way to say whether we have a well-defined outcome or not. Our lack of knowledge is expressed by the largest entropy value.

Von Neumann entropy

Using the density matrix ρ , we define the quantum mechanical equivalent of the classical entropy as

$$S = -\text{Tr}[\rho \log_2 \rho].$$

This is the so-called Von Neumann entropy. How did we arrive at this expression?

And why can't we use the classical definition for a quantum mechanical system as well?

The density matrix again

Let us assume we are studying specific system A . The density matrix, using an ONB basis ψ_j , is defined as

$$\rho_A = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j| = \sum_j p_j |\psi_j\rangle\langle\psi_j|,$$

and the eigenvalues λ_j are the probabilities (or overlap coefficients) of being in a specific state ψ_j .

Note that the density matrix/operator is a semi-positive definite matrix.

Linking with a new expression for the entropy

Using a unitary transformation $\mathbf{U}$ we can transform the density matrix into a diagonal matrix $\mathbf{D}_a$ where the eigenvalues are the above mentioned probabilities (overlap coefficients squared). We can then define

$$\mathbf{D}_A = \mathbf{U}^\dagger \rho_A \mathbf{U} = \begin{bmatrix} p_0 & 0 & 0 & \dots & 0 \\ 0 & p_1 & 0 & \dots & 0 \\ 0 & 0 & p_2 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & p_{n-1} \end{bmatrix} .$$

Von Neumann entropy

If we now construct the following object

$$\rho_A \log_2 \rho_A,$$

and perform a unitary transformation, recalling that

$$\mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U} = \mathbf{I},$$

$$\mathbf{U}^\dagger \rho_A \log_2 \rho_A \mathbf{U} = \mathbf{U}^\dagger \rho_A \mathbf{I} \log_2 \rho_A \mathbf{U} = \mathbf{D}_A \mathbf{U}^\dagger \log_2 \rho_A \mathbf{U},$$

and defining

$$\mathbf{U}^\dagger \log_2 \rho_A \mathbf{U} = \log_2 \mathbf{D}_A,$$

we have that

$$\mathbf{U}^\dagger \rho_A \log_2 \rho_A \mathbf{U} = \mathbf{D}_A \log_2 \mathbf{D}_A.$$

$$\log_2 \mathbf{D}_A$$

We have that

$$\log_2 \mathbf{D}_A = \begin{bmatrix} \log_2 p_0 & 0 & 0 & \dots & 0 \\ 0 & \log_2 p_1 & 0 & \dots & 0 \\ 0 & 0 & \log_2 p_2 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & \log_2 p_{n-1} \end{bmatrix}.$$

Entropy and traces

If we take the trace of

$$\mathrm{Tr}[\mathbf{U}^\dagger \rho_A \log_2 \rho_A \mathbf{U}] = \mathrm{Tr}[\mathbf{D}_A \log_2 \mathbf{D}_A],$$

which is nothing but

$$\mathrm{Tr}[\mathbf{D}_A \log_2 \mathbf{D}_A] = \sum_i p_i \log_2 p_i,$$

which allows us to define the general Von-Neumann entropy as

$$S_A = -\mathrm{Tr}[\rho_A \log_2 \rho_A].$$

Performing a unitary transformation gives the expression we know from classical information theory.

What next?

Before we proceed we are now going to study the entropy of a system relevant for project 1. We will compute the entropy of the ground state and the entropy of a subsystem by tracing out (as done previously) the density matrix of ground state.

The system is a two-qubit and we will use this later to rewrite the Hamiltonian matrix in terms of various Pauli-matrices.

Two-qubit system and calculation of density matrices and project 1

This part is best seen using the jupyter-notebook.

This system can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = [1 \ 0]^T \quad |1\rangle_{A,B} = [0 \ 1]^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry. This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \ 0 \ 0 \ 0]^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \ 1 \ 0 \ 0]^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \ 0 \ 1 \ 0]^T,$$

and finally

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \ 0 \ 0 \ 1]^T.$$

Schmidt decomposition

A way to study entanglement is through the so-called Schmidt decomposition, which is essentially an application of the singular-value decomposition.

Pure states and Schmidt decomposition

The Schmidt decomposition allows us to define a pure state in a bipartite Hilbert space composed of systems A and B as

$$|\psi\rangle = \sum_{i=0}^{d-1} \sigma_i |i\rangle_A |i\rangle_B,$$

where the amplitudes σ_i are real and positive and their squared values sum up to one, $\sum_i \sigma_i^2 = 1$. The states $|i\rangle_A$ and $|i\rangle_B$ form orthonormal bases for systems A and B respectively, the amplitudes λ_i are the so-called Schmidt coefficients and the Schmidt rank d is equal to the number of Schmidt coefficients and is smaller or equal to the minimum dimensionality of system A and system B , that is $d \leq \min(\dim(A), \dim(B))$.

Proof of Schmidt decomposition

The proof for the above decomposition is based on the singular-value decomposition. To see this, assume that we have two orthonormal bases sets for systems A and B , respectively. That is we have two ONBs $|i\rangle_A$ and $|j\rangle_B$. We can always construct a product state (a pure state) as

$$|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B,$$

where the coefficients c_{ij} are the overlap coefficients which belong to a matrix $\mathbf{C}$.

Further parts of proof

If we now assume that the dimensionalities of the two subsystems A and B are the same d , we can always rewrite the matrix $\mathbf{C}$ in terms of a singular-value decomposition with unitary/orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$ of dimension $d \times d$ and a matrix $\mathbf{\Sigma}$ which contains the (diagonal) singular values $\sigma_0 \leq \sigma_1 \leq \dots 0$ as

$$\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger.$$

SVD parts in proof

This means we can rewrite the coefficients c_{ij} in terms of the singular-value decomposition

$$c_{ij} = \sum_k u_{ik} \sigma_k v_{kj},$$

and inserting this in the definition of the pure state $|\psi\rangle$ we have

$$|\psi\rangle = \sum_{ij} \left(\sum_k u_{ik} \sigma_k v_{kj} \right) |i\rangle_A |j\rangle_B.$$

Slight rewrite

We rewrite the last equation as

$$|\psi\rangle = \sum_k \sigma_k \left(\sum_i u_{ik} |i\rangle_A \right) \otimes \left(\sum_j v_{kj} |j\rangle_B \right),$$

which we identify simply as, since the matrices $\mathbf{U}$ and $\mathbf{V}$ represent unitary transformations,

$$|\psi\rangle = \sum_k \sigma_k |k\rangle_A |k\rangle_B.$$

Different dimensionalities

It is straight forward to prove this relation in case systems A and B have different dimensionalities. Once we know the Schmidt decomposition of a state, we can immediately say whether it is entangled or not. If a state ψ is entangled, then its Schmidt decomposition has more than one term. Stated differently, the state is entangled if the so-called Schmidt rank is greater than one. There is another important property of the Schmidt decomposition which is related to the properties of the density matrices and their trace operations and the entropies. In order to introduce these concepts let us look at the two-qubit Hamiltonian described here.

Rewriting the above Hamiltonian as a set of Pauli-gates

Last week we discussed how to rewrite a one-qubit Hamiltonian in terms of various one-qubit gates. This forms our first Hamiltonian of project 1. We will move on to a two-qubit system. But first a short reminder about specific two-qubit gates.

Two-Qubit Gates

A two-qubit gate is a physical action that is applied to two qubits. It can be represented by a matrix U from the group $SU(4)$. One important type of two-qubit gates are controlled gates, which work as follows: Suppose U is a single-qubit gate. A controlled- U gate (CU) acts on two qubits: a control qubit $|x\rangle$ and a target qubit $|y\rangle$. The controlled- U gate applies the identity I or the single-qubit gate U to the target qubit if the control gate is in the zero state $|0\rangle$ or the one state $|1\rangle$, respectively.

Control qubit

The control qubit is not acted upon. This can be represented as follows if

$$CU|xy\rangle = |xy\rangle \text{ if } |x\rangle = |0\rangle.$$

In matrix form

It is easier to see in a matrix form. It can be written in matrix form by writing it as a superposition of the two possible cases, each written as a simple tensor product

$$CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{bmatrix}.$$

CNOT gate

One of the most fundamental controlled gates is the CNOT gate. It is defined as the controlled- X gate CX . It can be written in matrix form as follows:

$$\text{CNOT} = CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

CX gate

It changes, when operating on a two-qubit state where the first qubit is the control qubit and the second qubit is the target qubit, the states (check this)

$$\text{CX}|00\rangle = |00\rangle,$$

$$\text{CX}|10\rangle = |11\rangle,$$

$$\text{CX}|01\rangle = |01\rangle,$$

$$\text{CX}|11\rangle = |10\rangle,$$

which you can easily see by simply multiplying the above matrix with any of the above states.

Swap gate

A widely used two-qubit gate that goes beyond the simple controlled function is the SWAP gate. It swaps the states of the two qubits it acts upon

$$\text{SWAP}|xy\rangle = |yx\rangle.$$

and has the following matrix form

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Reminder from last week

We rewrote the one-qubit Hamiltonian

$$\mathbf{H} = \begin{bmatrix} \epsilon_1 + V_{11} & V_{12} \\ V_{21} & \epsilon_2 - V_{22} \end{bmatrix},$$

as (defining $V_{12} = V_{21} = V_x$)

$$\mathbf{H} = \frac{\epsilon_1 + \epsilon_2}{2} \mathbf{I} + \frac{\epsilon_1 - \epsilon_2}{2} \mathbf{Z} + \frac{V_{11} + V_{22}}{2} \mathbf{I} + \frac{V_{11} - V_{22}}{2} \mathbf{Z} + V_x \mathbf{X}.$$

We will now rewrite our first two-qubit Hamiltonian along the same lines.

Two-qubit Hamiltonian

We start again with

$$\mathbf{H} = \begin{bmatrix} \epsilon_1 + V_z & 0 & 0 & V_x \\ 0 & \epsilon_2 - V_z & V_x & 0 \\ 0 & H_x & \epsilon_3 - V_z & 0 \\ H_x & 0 & 0 & \epsilon_4 + V_z \end{bmatrix}.$$

This Hamiltonian, as discussed in the first exercise can be rewritten in terms of various one-qubit matrices.

Definitions

We define

$$\epsilon_{II} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4}{4},$$

$$\epsilon_{ZI} = \frac{\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{IZ} = \frac{\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{ZZ} = \frac{\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4}{4}.$$

Ex1: Two-qubit Hamiltonian

This exercise is also part of the first project. Rewrite the Hamiltonian in term of various Pauli matrices and show that you can rewrite the Hamiltonian as

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{H}_I,$$

with

$$\mathbf{H}_0 = \epsilon_{II} \mathbf{I} \otimes \mathbf{I} + \epsilon_{ZI} \mathbf{Z} \otimes \mathbf{I} + \epsilon_{IZ} \mathbf{I} \otimes \mathbf{Z} + \epsilon_{ZZ} \mathbf{Z} \otimes \mathbf{Z}$$

and

$$\mathbf{H}_I = V_x \mathbf{X} \otimes \mathbf{X} + V_z \mathbf{Z} \otimes \mathbf{Z}$$

Ex2: Two-qubit Hamiltonian and entropy

Use the Hamiltonian for the two-qubit example to find the eigenpairs as functions of the interaction strength λ and study the final eigenvectors as functions of the admixture of the original basis states. Discuss the results as functions of the parameter λ and compute the von Neumann entropy and discuss the results. You will need to calculate the entropy of the subsystems A or B .